Harshit Mathur

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GitHub | Kaggle | Codeforces

Education

Indian Institute of Technology (ISM) Dhanbad — Expected May 2027
Bachelor of Technology in Electronics and Communication Engineering (ECE) — *Current CGPA - 8.91/10*

- **Relevant Coursework:** Linear Algebra, Probability and Statistics, Calculus, Data Structures & Algorithms.

Work Experience

ML Research Intern, IIT Roorkee — May 2025 - July 2025

- Spearheading development of an ***unsupervised vision system*** utilizing ground-based visual data for proactive, ***real-time*** landslide early warning.
- Architecting scalable deep learning pipelines and pioneering ***domain generalization*** strategies to adapt ***optical flow*** models from aerial imagery to ground-level visual streams, bolstering system reliability and accuracy.

Projects

AI-Powered Agent for Authenticated News Curation and Publishing | PyTorch, NLP | [*Code*]

- Built an AI-driven system for autonomous news retrieval, verification, summarization, and publishing.
- Used a ***BERT***-base model for ***semantic similarity*** analysis to cross-verify claims.
- Leveraged ***Gemini*** for abstractive summarization while preserving ***factual integrity***.
- Automated ***AI-generated*** imagery using open-source image generation platform for enhanced readability.
- Developed an ***end-to-end*** pipeline for web crawling, fact validation, summarization, ***SEO***, and publishing.

Data-Centric Land Cover Classification - BMVC 2024 | Hugging-Face, Lightning, Hydra | [*Code*]

- Designed a novel ranking strategy to estimate label fidelity across samples, enabling noise-robust training.
- Leveraged synthetic data for model training using ***Lightning-Hydra*** Template, simulating real-world data.
- Benchmarked ***DeepLabv3+*** with ***ResNet152*** backbone, achieving ***0.7331 Val Mean IoU***.
- Employed ***Voxel51*** for inference visualization, analysis, and high-quality dataset curation to enhance training.
- Submitted ***75+*** model variants, including ***Grounding SAM, Grounding DINO, UNet, Mask2Former***, achieving a ***Kendall Tau*** score of ***0.4635***.

Technical Skills

Languages & Core Tools: C/C++, Python, Git, Github, VS Code, Neptune, Voxel51

Frameworks: PyTorch, Lightning, Hydra, Hugging Face, LangChain, LangGraph

Libraries: Hugging Face, OpenCV, timm, segmentation-models-pytorch, Scikit-learn, Librosa

Concepts: Object-Oriented Programming, Operating System, SQL, Computer Vision, Semantic Segmentation, CNNs, Natural Language Processing, Transformers, LLMs, Transfer Learning, RAG

Achievements

- **2nd Runner-Up (out of 445 teams)** : ***SHL*** Hiring Assessment Challenge hosted on ***Kaggle***.
- **2nd Runner-Up** : ***AGGLOMERATION 2.0***, hosted by CSE Society, ***AI of GOD 3.0***, hosted by MnC Department, IIT Dhanbad.
- **Codeforces :** ***Specialist*** (Max Rating : 1461); **Codechef :** ***3-Star*** (Max Rating : 1694); Solved ***500+*** problems.
- **Winners** - ***Parakram 2024*** (annual sports fest); **1st Runner-up** - ***Parakram 2025*** & ***Dhanbad District Tournament*** - As Volleyball team captain.

Social Engagements

- **Club Member**: Machine Learning Division, Data and Software Technology Club, IIT Dhanbad.
- **Coordinator**: Volleyball Club
- **Contingent Member**: Inter IIT Sports Meet 2023 (IIT Bombay), 2024 (IIT Kanpur), 2025 (IIT Hyderabad), Spardha '24 - IIT BHU Sports Fest , Shaurya 23' - IIT KGP Sports Fest.